

Smart Security Safe System



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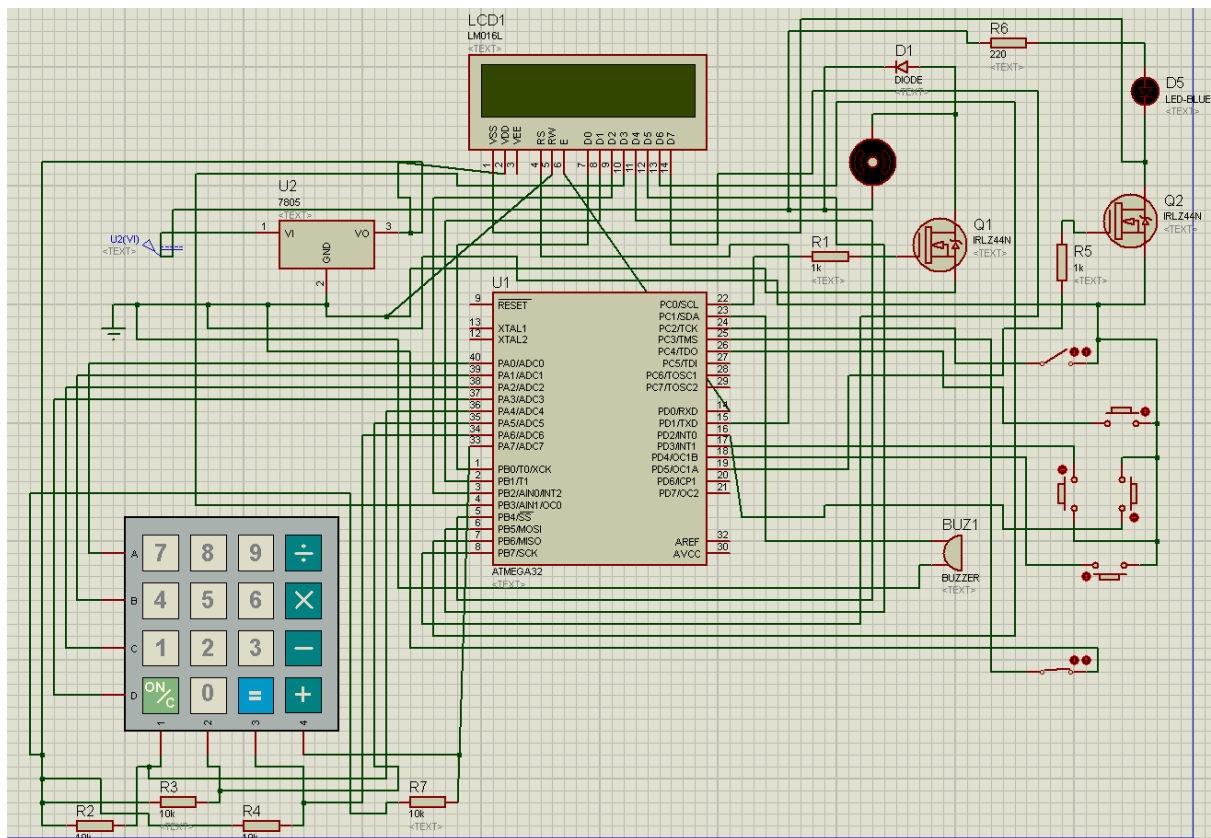
Project Overview

This project is a smart electronic safe system designed to provide multiple layers of security and a user-friendly control interface. The safe can be operated using either a password or a physical key for authentication and security management.

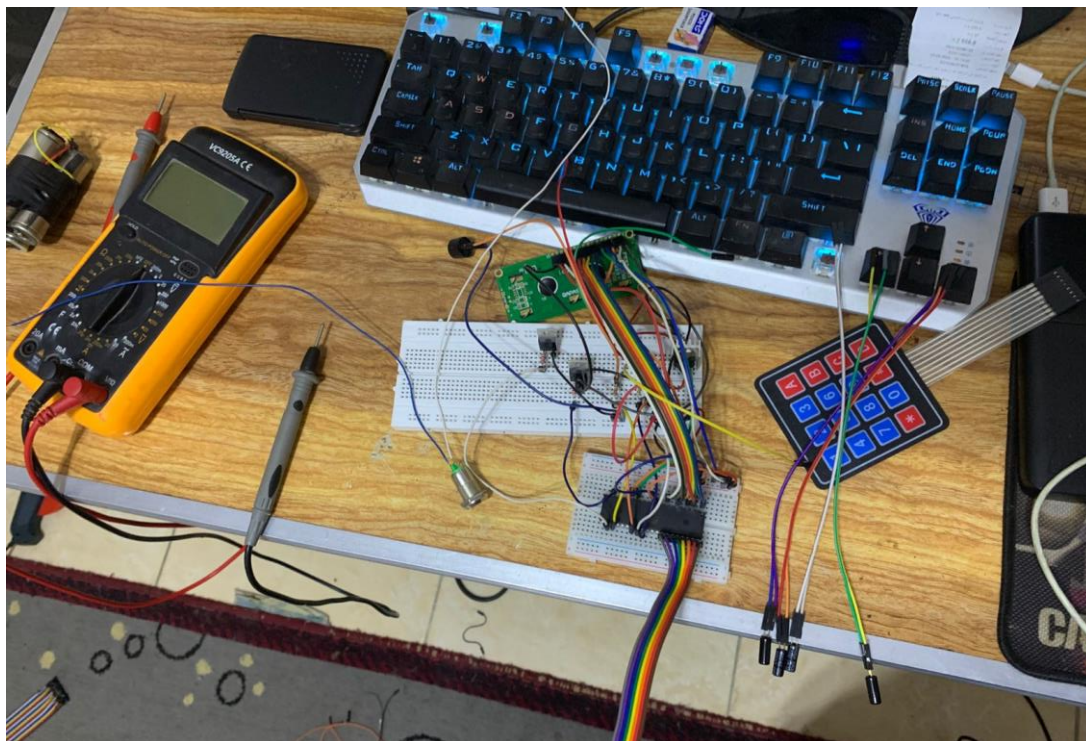
Main Features

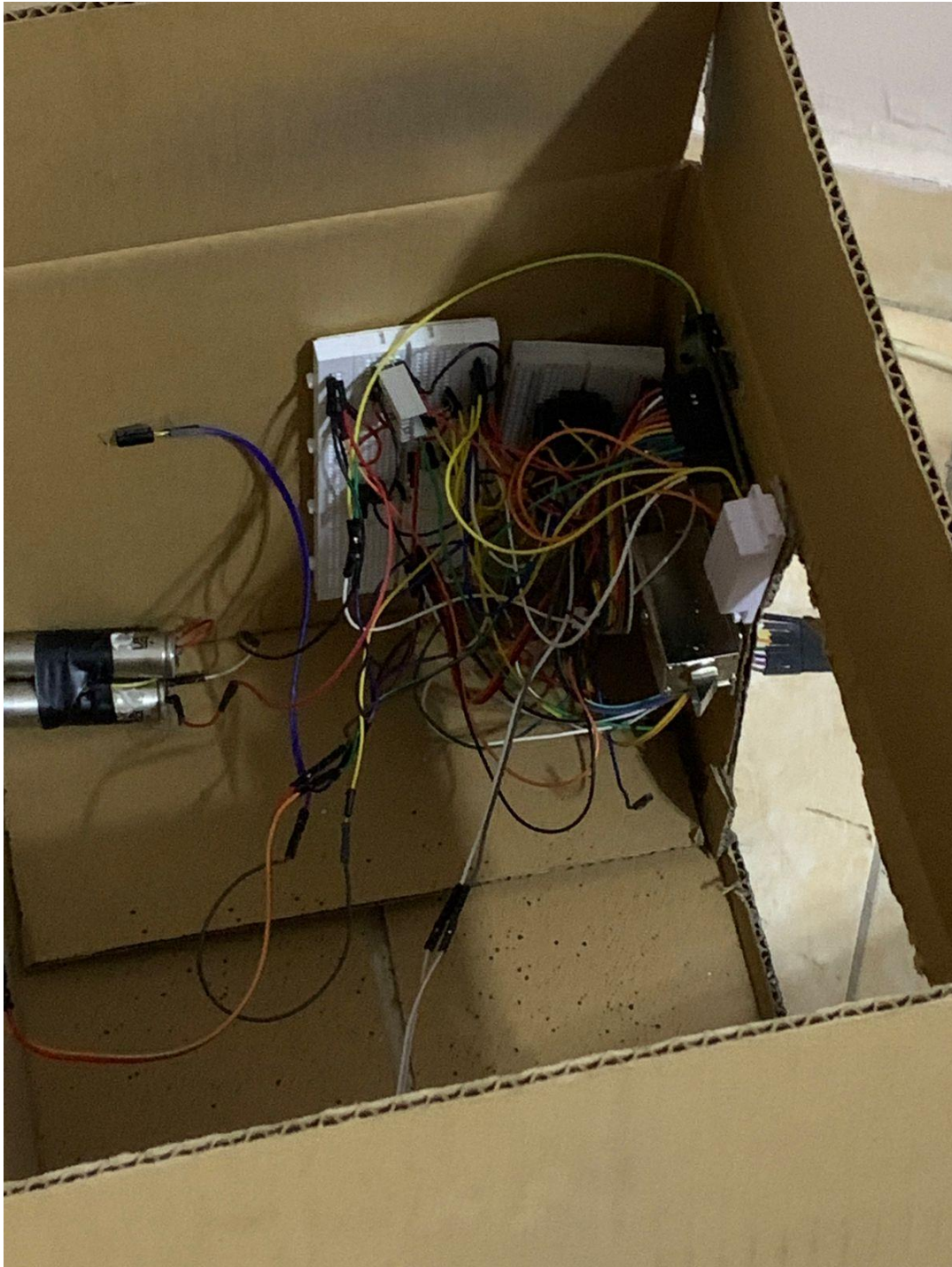
- Open the safe using a password or a physical key.
- Change the password using either password authentication or the physical key.
- Enable or disable transportation and anti-theft protection mode using either the password or the physical key.
- Navigate easily through a menu-based LCD interface.
- The password is securely stored in EEPROM memory to preserve data even after power loss.

System Overview



Proteus Design





Hardware Implementation

LCD and User Interface

The system uses an LCD display with several smart power-saving and usability features:

- If the user stays on the main menu without pressing any keypad button for three menu cycles, the LCD backlight turns off automatically to save power.
- Pressing any keypad button turns the LCD backlight on again
- If the user stays inside any option (such as opening the safe, changing the password, or protection settings) for 20 seconds without interaction, the system automatically returns to the main menu.
- Password characters are hidden as * symbols while typing.
- Pressing * toggles password visibility between hidden stars and visible numbers.

- Pressing # deletes the last entered digit from the password input.
- Pressing B at any time returns the user to the main menu.
- The system provides interactive sound feedback when buttons are pressed or when an incorrect password is entered.

Security System

The safe includes multiple protection mechanisms to prevent unauthorized access or theft:

- Four SW-520D vibration sensors are installed on the four sides of the safe to detect movement, carrying attempts, or breaking attempts.
- A magnetic switch is installed on the safe door to detect unauthorized opening.
- If the safe door is opened without using the correct password or key, the alarm is activated immediately.
- Transportation and anti-theft protection mode can be enabled or disabled using either the password or the physical key.
- The four vibration sensors are distributed on all four sides of the safe for full coverage detection.

Alarm System

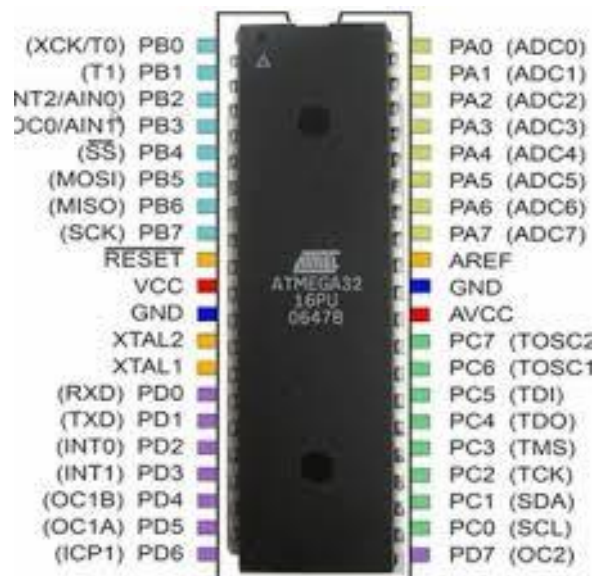
The alarm is triggered in the following situations:

- Entering the wrong password three consecutive times.
- Detecting vibration or movement while protection mode is enabled.
- Detecting unauthorized door opening using the magnetic switch.

Once activated, the alarm can only be turned off using the physical key for maximum security.

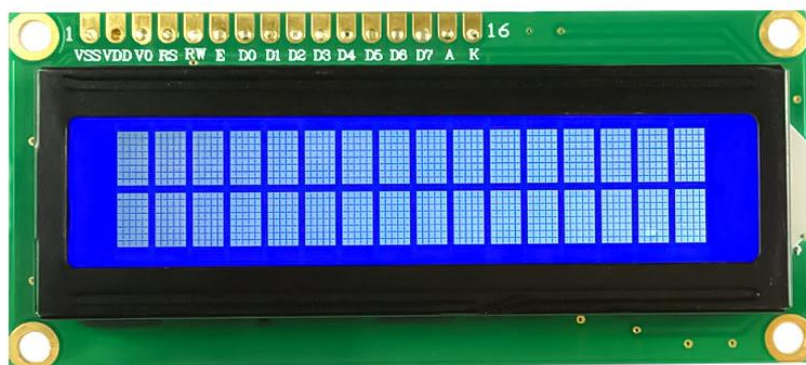
Hardware Components

1:



ATmega32 Microcontroller

2:



16x2 LCD Display

3:



4x4 Keypad

4:



4x SW-520D Vibration Sensors

5:



Magnetic Door Switch

6:



Buzzer

7:



12V DC Lock

8:



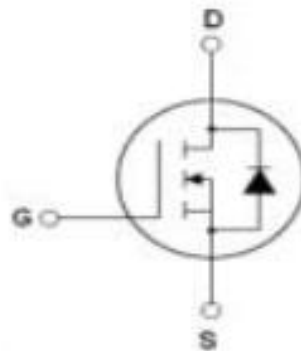
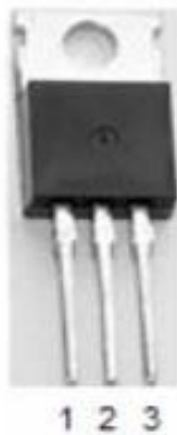
12V Battery

9:



Key Switch

10:



pin 1, Gate
2, Drain
3, Source

2x IRLZ44Z MOSFET N-Channel

11:



4x 10K Resistance

12:



2x 1K Resistance

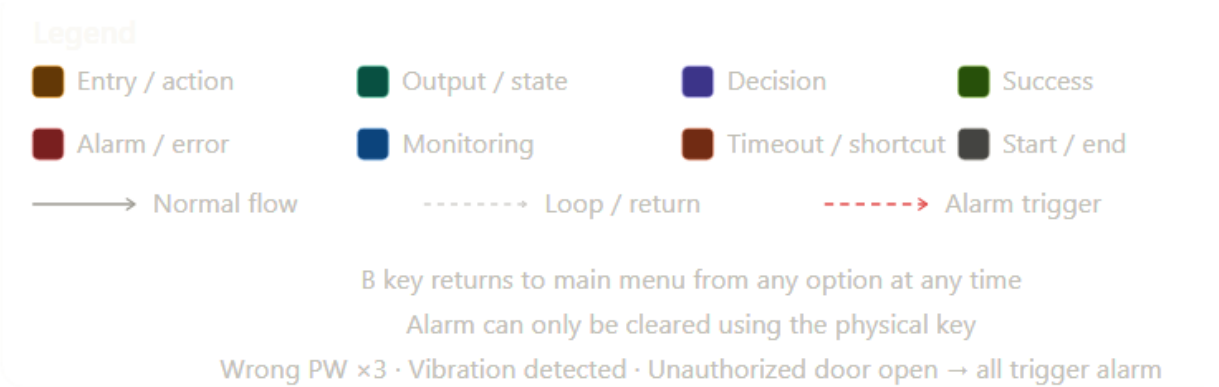
Software and Tools Used

- Atmel Studio – Firmware development and Embedded C programming
- Proteus Design Suite – Circuit design and system Simulation
- Extreme Burner – Flashing firmware to the Microcontroller
- Zadig – USB driver installation for programming tools (AVR/USB interface)

System Behavior Summary

- Secure authentication using password or physical key.
- Automatic LCD power saving system.
- Automatic return to main menu after inactivity.
- Hidden password input with toggle visibility feature.
- Multi-layer alarm triggering system.
- Full anti-theft and transportation protection Mode





System Flowchart

Challenges Faced

- Stabilizing readings from vibration sensors (SW-520D).
- Designing reliable menu navigation system.
- Implementing accurate timing for auto-return and LCD timeout.
- Managing alarm logic without false triggers.
- Synchronizing multiple security inputs (sensors, keypad, key switch).
- Designing the system to continuously monitor security sensors in parallel with menu navigation and password handling without blocking critical alarm functionality.

Future Improvements

- Add fingerprint authentication.
- Add GSM/Wi-Fi notification system.
- Store access logs in EEPROM or external memory.
- Add mobile application control.
- Integrate the safe with an external security/alarm system capable of automatically contacting authorities in case of theft attempts, forced opening, or unauthorized movement.
- Add a hidden camera module that captures images when an incorrect password is entered or when any security alarm is triggered.
- Add a GPS tracking system capable of transmitting the safe location to nearby smartphones through Bluetooth communication if the safe is stolen

